MD IMON HOSSAIN

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PROFESSIONAL SUMMARY

Analytical Chemist with graduate-level research and 2+ years of multi-sector laboratory experience across pharmaceutical, environmental, and consumer product testing. Skilled in HPLC, UHPLC, LC-MS, GC-MS, and ICP-MS for method development, validation, and troubleshooting in cGMP/GLP-regulated environments. Experienced in regulated QC testing (ISO/GMP), stability studies, and structural confirmation of complex matrices, rigorous method validation and data integrity. Authorized to work in the US, no sponsorships are required.

EDUCATION

Master of Science in Chemistry — January 2025 - December 2026 (Est.)
Western Illinois University, IL, USA — GPA: 4.00/4.00

Bachelor of Science in Chemistry — March 2018 - January 2024
Pabna University of Science and Technology, Bangladesh — GPA: 3.04/4.00

TECHNICAL SKILLS

- **Chromatography & Spectroscopy:** HPLC, UHPLC, GC-MS, LC-ESI-QTOF-MS, Tandem MS, UV-Vis Spectrophotometry, ICP-OES, ICP-MS, HRMS, FTIR, HILIC, Reversed-Phase LC, NMR
- **Laboratory Operations:** Method Development, Method Validation, Sample Preparation, Stability Studies, Wet Chemistry, Calibration, Instrument Troubleshooting
- **Quality & Compliance:** cGMP, GLP, SOPs, Data Integrity, OOS, CAPA, Root Cause Analysis, QA/QC documentation
- **Software & Tools:** MassHunter, Delta, ChemDraw, OMNIC, Gaussian, Microsoft Word/Excel/PowerPoint/Teams

PROFESSIONAL EXPERIENCE

Research Assistant, Department of Chemistry | Western Illinois University — **May 2025 - Present**

- Developed and validated high-resolution LC-MS methods for quantitative analysis of complex multi-analyte mixtures, delivering regulatory-compliant QA/QC data and supporting OOS/OOT investigation readiness.
- Designed and executed 64-week stability studies (-80°C to +80°C) on finished consumer products, applying ICH-aligned testing principles.
- Quantified trace-level analytes in personal care and consumer product matrices via UHPLC-DAD and ESI-TOFMS, resolving structural isomers relevant to product safety and impurity profiling.
- Optimized extraction and sample preparation protocols for cannabinoid isolation from complex matrices, improving recovery and reproducibility. Analyzed chromatographic and spectrometric data to support method validation.
- Led a 7-member research team on cross-functional projects, ensuring on-time execution and QA/QC-ready documentation.

Teaching Assistant, Department of Chemistry | Western Illinois University — **August 2025 - Present**

- Provided instruction in chromatography, wet-lab operations, and analytical data interpretation, enforcing GLP practices.
- Developed training materials and technical guidance to support process consistency across lab sessions.
- Evaluated performance and improved accuracy, compliance, and productivity for 50+ students per semester.

Intern Analytical Chemist | Intertek — **November 2023 - December 2023**

- Executed ISO-compliant QC testing using HPLC-MS, GC-MS, ICP-OES, and ICP-MS to support consumer and environmental safety evaluation.
- Conducted qualitative and quantitative analysis of pH, formaldehyde, heavy metals, bisphenols, dyes, PFAS, phthalates, APEO, and PAHs across garment, water, and effluent treatment plant (ETP) samples.
- Executed raw material testing and method validation of finished products per SOPs and ISO guidelines.
- Managed high-throughput testing (100+ samples/day), maintaining data integrity, statistical review, and timely reporting
- Supported deviation investigations and continuous improvement initiatives using 5S, PDCA, and Kaizen methodologies.

PUBLICATIONS

- *Quantitative analysis of epimeric and Δ^8/Δ^9 positional isomers of synthetic THC analogs and natural cannabinoids in formulated products by UHPLC-DAD with optional ESI/TOFMS confirmation.* Journal of Liquid Chromatography & Related Technologies, 1-14. https://doi.org/10.1080/10826076.2026.2636540
- *High-Throughput LC-MS/MS Quantification of Eighteen Cannabinoids in Hemp Flowers with Baseline Separation of Structural Isomers.* Molecules 2026, 31, 1684. https://doi.org/10.3390/molecules31101684
- *Efficacy of phytochemicals derived from Artocarpus heterophyllus (Jackfruit) as inhibitors against NS2B/NS3 protease of dengue virus: an in-silico investigation.* Scientific Reports (2026). https://doi.org/10.1038/s41598-026-38726-x